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RIM-reference ideal method in multicriteria decision making

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ABSTRACT

Multiple criteria decision making methods provide a useful tool to support decision making. To date, different methods have been developed using different approaches. In particular, the conception used by the TOPSIS and VIKOR methods identifies the best alternative based on the positive and the negative ideal solutions.

In this paper, a new method based on the concept of “ideal solution” is presented as a possible variant of the aforementioned. However, in our case these values can vary between the maximum value and the minimum value; this does not occur in the VIKOR and TOPSIS methods in which these values are the extreme values. Moreover, our method is characterized to be independent of the type of data. Moreover, RIM does not present Rank Reversal, an aspect that is not present in other multiple criteria decision making methods.

1. Introduction

It is very common to deal with situations where a set of alternatives exists, which are assessed according to certain criteria and that we wish to choose the best among them; this is, in essence, a decision making problem. However, such criteria are usually in conflict and there may be no solution satisfying all criteria simultaneously [24,43].

One very useful tool for solving the aforementioned situation is to use Multiple Criteria Decision Analysis (MCDA), which helps the decision-makers during the decision making process. The methods that comprise it allow to approach the problem in an orderly way, facilitating the consensus of the final decision and the treatment of the large amount of information which is usually in different magnitudes of measurement and meanings.

Different MCDA methods have been developed in the literature which use different paradigms and conceptions. They can be classified as follows:

- The American School, which has as its conception the use of a utility function to obtain the information aggregation of the different criteria; for example, the sum average weighted (SAW), the Analytic Hierarchy Process (AHP) [33], the Simple Multiattribute Rating Technique (SMART) [15], and others.
- The European School, represented mainly by the French School, which bases its conception under the basic principle of establishing a preference relationship between alternatives; for example, the ELimination and Choice Expressing Reality (ELECTRE) [32] and the Preference Ranking Organization Method for Enrichment of Evaluations (PROMETHEE) [3], with their respective variants.

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- Other methods which use different conceptions such as: The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) [20]; the Multicriteria Optimization and Compromise Solution (VIKOR) [29]; the Closed Procedures near Reference Situations (ZAPROS) [25]; the lexicographical method [16], among others.

In this way, it is possible to consider a subgroup that involves aspects pertaining to costs and benefits. In particular, the TOPSIS and VIKOR methods are rational and use the principle of identifying the Positive Ideal solution (PIS) and the Negative Ideal Solution (NIS), based on the maximum value and/or the minimum value accordingly. The objective of these methods is to obtain the alternatives that are nearest to the PIS and as far as possible from the NIS. However, these methods introduce different forms of aggregation functions for ranking and different normalizations to equalize the possibly distinct ranges of each criterion function. Therefore, the VIKOR method considers the relative importance of all the criteria, and a balance between total and individual satisfaction, whereas the TOPSIS method introduces an aggregating function based on PIS and NIS points without considering their relative importance. In both methods the importance weights are incorporated into the comparison procedures.

If we query the SCOPUS database, a total of 2844 documents appear with the input TOPSIS. A number of papers are devoted to fuzzy extension of the TOPSIS method in the literature [14,38]. There are many applications of TOPSIS in the literature and particularly in the case of fuzzy TOPSIS. For instance, Cebi and Kahraman [5] developed a group decision support system to obtain the optimal selection of the location for emergency services. Chen [9] extended TOPSIS to the fuzzy environment and gave a numerical example of analysis selection for a software company. Tsaur et al. [37] applied fuzzy set theory to evaluate the quality of airline services. Chu [12] presented a fuzzy TOPSIS model under group decisions for solving a facility location selection problem. Chu and Lin [13] proposed the fuzzy TOPSIS method for robot selection. Jahanshahloo [23] used the concept of α-cuts to normalize fuzzy numbers. Chen et al. [10] presented a fuzzy TOPSIS approach to deal with the supplier selection problem in a supply chain system. Li [26] gave a comparative analysis of the compromise ratio method and the extended fuzzy TOPSIS method and highlighted their similarities and differences with an illustrative numerical example. Benitez et al. [2] presented a fuzzy TOPSIS method for measuring the quality of service in the hotel industry. García-Cascales et al. [17,19] studied the maintenance management problem in an engine factory and the evaluation of photovoltaic cells, respectively. Maity and Chakraborty [28] proposed the Fuzzy TOPSIS method to select the most suitable abrasive material for a grinding wheel. Yang and Hung [40] proposed to use TOPSIS and fuzzy TOPSIS methods for a plant layout design problem. It is, however, possible to work with interval values in the TOPSIS method as in [42].

Nevertheless, only 389 results were returned for the input VIKOR, 310 of which were also related to TOPSIS. Due to its characteristics and capabilities, the VIKOR method has been increasingly used in recent years. In a recent work, Anojkumar et al. [1], made a comparative analysis of both methods for pipe material selection in the sugar industry. Chang and Lin [7] studied hospital service evaluation in Taiwan. Shanian [36] used these approaches for a material selection problem in high safety requirements in structural elements of the aerospace and nuclear industries. Chakraborty and Chatterjeeb [6] applied VIKOR, TOPSIS and PROMETHEE methods for a material selection problem. Chiu et al. [11] combined DANP with VIKOR to improve e-store business. Jahan et al. [22] used the VIKOR method for a material selection problem with interval numbers. Opricovic and Tzeng [30] provided a detailed comparison of TOPSIS and VIKOR and explained that the compromise solution (VIKOR) gives a maximum group utility of the group majority and a minimum individual regret of the opponent. Wu et al. [39] developed a hybrid fuzzy model application for the innovation capital indicator assessment of Taiwanese Universities using FAHP and VIKOR. In a similar context, Zolfani and Ghadikolaei [44] obtained the ranking of private universities in Iran using VIKOR. Ilangkumaran and Kumanan [21] applied VIKOR to select a suitable maintenance strategy for the frame unit of a textile spinning mill, while Büyüközkan and Ruan [4] used the same method to measure the performance of enterprise resource planning software products. Sanayei et al. [35] proposed a group decision making process for supplier selection with fuzzy VIKOR. San Cristóbal [34] proposed the VIKOR method for the selection of a renewable energy project in Spain and Chang [8] has studied the water quality conditions in a watershed. Assessment of health-care waste disposal methods using a VIKOR-based fuzzy multi-criteria decision making method can be seen in [27]. The main advantage of the VIKOR method is that it introduces the multi-criteria ranking index based on the particular measure of “closeness” to the ideal solution [31]. Chiu et al. [11] combines DANP with VIKOR to evaluate products and services from e-stores. Yang et al. [41], applied VIKOR with DEMATEL and ANP to security risk control.

The conception of the PIS and the NIS is an element of great importance for some MCDA. In these methods it is only necessary to determine the distance from each alternative to the ideal solutions. The alternatives are then ordered according to this distance. However, the procedure used to characterize the positive and negative ideal values consists in associating them to the maximum or minimum values, respectively and accordingly, depending if they are benefits or costs.

Such an approach is a drawback of these methods, because in practice, the ideal solution is not necessarily one of the extreme values, but may in fact be a value somewhere in between. For example, consider the case for selecting a driver for an entity. If the age is one of the criteria being assessed, the person wanted should be between 30 and 35 years old in the ideal case. Assume the age range of our candidates is between 23 and 60 years old; it is then evident that the PIS and the NIS do not have to be 23 nor 60 years old.

Another drawback present in the methods analyzed previously is that they are dependent on the data; this implies that when including a new alternative or only by modifying the data of one of the alternatives, it is then necessary to carry out the aggregation of the information for all the alternatives.

Based on the problem outlined previously, it becomes necessary to extend the concept of the MCDA, on the basis that the decision-makers define the reference ideal, and this can take values between the minimum and the maximum values of the range. Moreover, the proposed method allows operating with intervals, labels or simple values.

This paper is organized into five sections. The introduction presents the outline of MCDA, with particular emphasis on the TOPSIS and VIKOR methods, as well as the constraints that both present. The basic concepts to be used are defined in the second section, which briefly reviews the similarities and differences between TOPSIS and VIKOR. The third section describes the new method and the normalization function employed. Through an example we illustrate its practical use in the fourth section. Finally the conclusions of the work are presented.

2. MCDA with reference to ideal

In a general way, the different MCDA begins with a set of alternatives that should be evaluated taking into account a set of criteria. Through an algorithmic process that guides the decision-makers it is possible to obtain the best alternatives. It is also common that these methods use a weights vector to indicate the relative importance of each criterion; this allows us to establish a certain influence for each criterion and thus will affect the final result.

Among the many compensatory approaches of MCDA it is possible to consider a subgroup that involves costs and benefits aspects. One such approach is TOPSIS, with VIKOR being another. These approaches are employed in the literature for four main reasons:

a. TOPSIS logic is rational and understandable;
b. the computation processes are straightforward;
c. the concept permits the pursuit of the best alternatives for each criterion depicted in a simple mathematical form;
d. the importance weights are incorporated into the comparison procedures.

These ideas allow us to develop the new approach.

2.1. Basic concepts

Decision-making is the procedure to find the best alternative among a set of feasible alternatives. Decision-making problems considering several criteria are sometimes referred to as multi-criteria decision-making (MCDM) problems [20,24,43])

In the MCDM process, it is necessary to define a series of elements involved in the process, thus:

- Decision-makers are the entity responsible for the selection of a possible alternative.
- The alternatives are the possible actions to choose by the decision-makers.
- The criterion or attribute is the characteristic, parameter or reference point that is used to describe the alternatives' qualities.
- The valuation matrix represents the valuations of all the alternatives for each criterion, with the elements x_{ij} representing the evaluation or judgments of the alternative *Ai* for the criterion *Cj*. The weights are the measures that indicate the relative importance of the criteria for the decision-makers.

A MCDA problem with m alternatives and n criteria can be expressed in matrix format as follows:

$$X = \begin{array}{c} \\ A_1 \\ A_2 \\ \vdots \\ A_m \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} C_1 & C_2 & \dots & C_n \end{array} \\ \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \\ \begin{array}{cccc} w_1 & w_2 & \cdots & w_n \end{array} \end{array}$$

where $A_1, A_2, \dots, A_m$, are feasible alternatives, $C_1, C_2, \dots, C_n$, are evaluation criteria, x_{ij} is the performance rating of alternative A_i under criterion C_j, and w_j is the weight of criterion C_j.

Generally speaking, we can state that the selection process of the best alternative begins with the criteria set identification. Then, the decision-makers use the criteria set to evaluate each alternative, and from them obtain an array of judgments. Finally, the aggregation process allows the transformation of the valuation matrix into a vector that represents the result for each alternative (see Fig. 1).

2.2. TOPSIS vs. VIKOR

TOPSIS and VIKOR are well-known methods for classical MCDM that have been applied in different areas of research. Both of them deal with different types of data. The TOPSIS and VIKOR methods use the concept of identifying the best alternative from the PIS and the NIS, associating them to the maximum value and the minimum value, respectively, depending if the criterion is a benefit or a cost. These methods have in common:

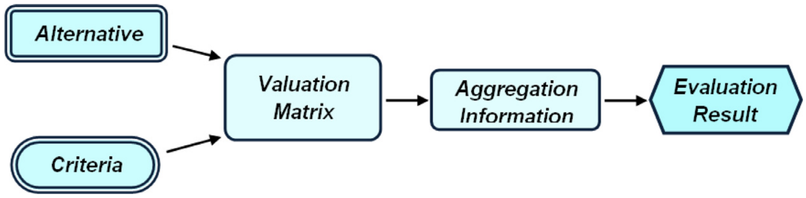

Fig. 1. Flow chart of the evaluation of alternatives.

Table 1
Differences between TOPSIS and VIKOR.

TOPSIS method	VIKOR method
• Normalization of the valuation matrix through: $r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{j=1}^{m} x_{ij}^2}}$ • Calculation of the separation measure to the PIS and the NIS. $S_j^+ = \sqrt{\sum_{i=1}^{n} (v_{ij} - v_i^+)^2}, S_j^- = \sqrt{\sum_{i=1}^{n} (v_{ij} - v_i^-)^2}$ Where: ■ $j = 1, 2, \ldots, m$; $i = 1, 2, \ldots, n$ ■ v_i^+ and v_i^-, represent the PIS and NIS, respectively. ■ v_{ij}, is the normalized and pondered valuation for the alternatives in the respective criteria. • Calculation of the relative index to the ideal solution $R_j = \frac{S_j^-}{S_j^+ + S_j^-}$, where: $0 < R_j < 1$, $j = 1, 2, \ldots, m$ • To order the alternatives starting from the relative index R_j.	• Calculations of the following indexes: $S_j = \sum_{i=1}^{n} \frac{w_i(f_i^+ - f_{ij})}{(f_i^+ - f_i^-)}$, $R_j = \max_i [\frac{w_i(f_i^+ - f_{ij})}{(f_i^+ - f_i^-)}]$ Where: ■ $j = 1, 2, \ldots, m$; $i = 1, 2, \ldots, n$ ■ f_i^+ and f_i^-, represent the PIS and NIS, respectively. ■ f_{ij}, is the valuation of the alternatives in function of the criteria. • Final index, $Q_j = v(\frac{S_j - S^+}{S^- - S^+}) + (1 - v)(\frac{R_j - R^+}{R^- - R^+})$ where, $S^+ = \min_j S_j$, $S^- = \max_j S_j$, $R^+ = \min_j R_j$, $R^- = \max_j R_j$ and $0 \leq v \leq 1$ • To order the alternatives.

- A valuation or judgments matrix.
- The normalization of the valuation matrix.
- A weights vector to indicate the relative importance among the criteria.
- The identification of the PIS and the NIS is associated to the maximum value and the minimum value, respectively and so corresponds. In the case of VIKOR this idea is incorporated into the norm.

On the other hand, these methods differ in the metric used to calculate the degree of grade of each alternative from the ideal solutions, as shown in Table 1.

As shown, these methods focus on the search for the ideal solution in the maximum value and/or the minimum value. Nevertheless a great diversity of decision problems exists and thus the ideal solution is not always strictly the maximum value and/or the minimum value, but rather the ideal solution is a value (or a set of values) that lies somewhere in between.

Starting from what has been outlined above, we propose the Reference Ideal Method (RIM), in which the reference ideal is a value or set of values that can be between the maximum value and the minimum value.

3. RIM-the Reference Ideal Method

As mentioned previously, the criteria for decision-makers constitute the reference point to assess the qualities of each alternative. However, these criteria do not necessarily have the same domains associated, nor the same range. For such reasons, the valuations emitted for each criterion must be normalized.

This normalization of the valuation matrix can be carried out in different ways depending on the MCDA. In our case, we consider doing the normalization relating to the reference ideal.

3.1. RIM normalization

To carry out the normalization process in RIM, it is first necessary to bear in mind that each criterion has associated a domain D belonging to a universe of discourse. It is also necessary to identify certain concepts, such as:

- *Range*. This is any interval, labels set or simple set of values that belong to a domain D.
- *Reference Ideal*. This is an interval, labels set or simple values that represents the maximum importance or relevance in a given Range.

On the basis of the concepts referred to, it is possible to say that the reference ideal is included in the range. Therefore, the reference ideal may vary among the extreme values of the range. Thus, the reference ideal can be any set between the minimum value and the maximum value or can be a point.

Starting from the established conditions, and continuing with the TOPSIS algorithm, we need to normalize the values of the decision matrix. For this, the distance to the reference ideal must be known and is expressed by the function:

$$d_{\min}(x,[C,D]) = \min(|x-C|,||x-D) \tag{1}$$

where x is the valuation for a given approach and the interval $[C,D]$ is the reference ideal.

Then, to carry out the normalization we consider operating based on the Range and the Reference Ideal, by the following function:

- $f : x \oplus [A,B] \oplus [C,D] \to [0,1]$, such that:

$$f(x,[A,B],[C,D]) = \begin{cases} 1 & si \quad x \in [C,D] \\ 1 - \frac{d_{\min}(x,[C,D])}{|A-C|} & si \quad x \in [A,C] \ \wedge \ A \neq C \\ 1 - \frac{d_{\min}(x,[C,D])}{|D-B|} & si \quad x \in [D,B] \ \wedge \ D \neq B \end{cases} \tag{2}$$

where:

- $[A,B]$ is the range that belongs to a universe of discourse.
- $[C,D]$ represents the Reference Ideal.
- $x \in [A,B]$.
- $[C,D] \subset [A,B]$

The function f allows to obtain a value that belongs to the unitary interval, which means that if the obtained value is equal to 1, then it coincides with the defined Reference Ideal and the more distant it is from the value of 1, then the more distant it is from the reference ideal, independently of the evaluated variables.

Example 1. Let us consider the age criterion, such that the Range is $t = [23,60]$, the Reference Ideal is the interval $s = [30,35]$, and the value to be normalized is $x = 25$, then, taking into account function f, the second branch of the function is applied, because $x \in [23,30]$, and it is obtained that:

$$d_{\min}(25,[30,35]) = \min(|25-30|,|25-35|) = \min(5,10) = 5$$
$$f_1(25,[23,60],[30,35]) = 1 - \frac{d_{\min}(25,[30,35])}{|23-30|} = 1 - \frac{5}{|-7|} = 1 - \frac{5}{7} = 1 - 0.714 = 0.286$$

3.2. Procedure to apply RIM

The RIM method is organized in the following way:

Step 1. Define the work context.

In this step the conditions in the work context are established, and for each criterion the following aspects are defined:

- The Range t_j.
- The Reference Ideal s_j. The reference ideal can be any set between the minimum value and the maximum value.
- The weight w_j associated to the criterion.

Step 2. Obtain the valuation matrix X, in correspondence with the defined criteria.

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix}$$

Step 3. Normalize the valuation matrix X with the reference ideal.

$$Y = \begin{pmatrix} f(x_{11},t_1,s_1) & f(x_{12},t_2,s_2) & \cdots & f(x_{1n},t_n,s_n) \\ f(x_{21},t_1,s_1) & f(x_{22},t_2,s_2) & \cdots & f(x_{2n},t_n,s_n) \\ \vdots & \vdots & \ddots & \vdots \\ f(x_{m1},t_1,s_1) & f(x_{m2},t_2,s_2) & \cdots & f(x_{mn},t_n,s_n) \end{pmatrix}$$

where, f is the function considered in (2).

Step 4. Calculate the weighted normalized matrix Y', through:

Table 2
Description of the work context.

Criteria	Meaning	Weights	Range	Reference ideal
C_1	Age	0.2262	[23, 60]	[30, 35]
C_2	Years of experience	0.2143	[0, 15]	more than 10
C_3	Quantity of sanctions	0.1786	[0, 10]	0
C_4	Elementary knowledge of mechanics	0.1429	Good, fair, bad	Good
C_5	Physical limitations	0.1190	Does not have, partial disability, total disability	Does not have
C_6	Emotional stability	0.1190	High, normal high, normal, normal low, low	High, normal high

Table 3
The valuation matrix.

Alternative	C1	C2	C3	C4	C5	C6
A	30	0	2	3	3	2
B	40	9	1	3	2	2
C	25	0	3	1	3	2
D	27	0	5	3	3	1
E	45	15	2	2	3	4

$$Y' = Y \otimes W \begin{pmatrix} y_{11} \cdot w_1 & y_{12} \cdot w_2 & \dots & y_{1n} \cdot w_n \\ y_{21} \cdot w_1 & y_{22} \cdot w_2 & \dots & y_{2n} \cdot w_n \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} \cdot w_1 & y_{m2} \cdot w_2 & \dots & y_{mn} \cdot w_n \end{pmatrix}$$

Step 5. Calculate the variation to the normalized reference ideal for each alternative A_{i}.

$$I_i^+ = \sqrt{\sum_{j=1}^{n} (y'_{ij} - w_j)^2}, \quad I_i^- = \sqrt{\sum_{j=1}^{n} (y'_{ij})^2} \quad \text{and}\, i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n$$

Taking the above into account, the vector that represents the reference ideal will be the vector $(1, 1, \dots, 1)$; also, the reference ideal would be weighted, then the reference ideal coincides with the vector of weight w.

Step 6. Calculate the relative index of each alternative A_{i}, through the expression:

$$R_i = \frac{I_i^-}{I_i^+ + I_i^-} \quad \text{where}\, 0 < R_i < 1, \quad i = 1, 2, \dots, m$$

Step 7. Rank the alternatives A_i, i=1,2,...,m in descending order. The alternatives that are at the top constitute the best solutions.

As can be seen, the RIM method allows making the information aggregation from a reference ideal.

4. A real world application of the proposed method

For the purposes of illustrating the RIM method, it was applied in a process of personnel selection to evaluate the best proposals among five candidates who aspired to occupy the position as a driver in an entity. For confidentiality reasons the name of the entity is withheld, as are the names of the candidates. For this reason the candidates will be identified by the letters $\{A, B, C, D, E\}$.

The definition of the work context is shown in Table 2

In this problem, the range of the criteria C_4, C_5 and C_6, are defined by linguistic labels. The linguistic labels are associated a crisp value that will be an integer value that is from n to 1 (better to worse) as can be seen in Table 3.

The normalized process is shown in the following.

With the aim of illustrating the use of the normalization function, we make the corresponding calculation for the alternative B and the criterion C_4. The parameters to use are: the range (Good, Fair, Bad) that is associated to the crisp values from N to 1, then $t_B = \{3, 2, 1\}$. The reference ideal is S_B={Good}, therefore $s_B = \{3\}$. Then when operating with the crisp values described in Table 3, we obtain the following calculations:

$$d_{\min}(3, \{3\}) = |3 - 3| = 0$$
$$f_1(3, \{3, 2, 1\}, \{3\}) = 1 - \frac{d_{\min}(3, \{3\})}{|3 - 1|} = 1 - \frac{0}{|2|} = 1 - \frac{0}{2} = 1$$

In a similar way we calculate the values corresponding to the criteria so that their ranges are linguistic labels. The numeric values are calculated in a similar manner to the example #1, collected in Table 4.

Table 4
Normalized valuation matrix.

Alternative	C1	C2	C3	C4	C5	C6
A	1	0	0.8	1	1	0.3333
B	0.8	0.9	0.9	1	0.5	0.3333
C	0.2857	0	0.7	0	1	0.3333
D	0.5714	0	0.5	1	1	0
E	0.6	1	0.8	0.5	1	1

Table 5
Weighted normalized matrix.

Alternative	C1	C2	C3	C4	C5	C6
A	0.22620	0	0.14288	0.14290	0.11900	0.03967
B	0.18096	0.19287	0.16074	0.14290	0.05950	0.03967
C	0.06463	0	0.12502	0	0.11900	0.03967
D	0.12926	0	0.08930	0.14290	0.11900	0
E	0.13572	0.21430	0.14288	0.07145	0.11900	0.11900

Table 6
Indexes calculation.

Alternative	I_i^+	I_i^-	R_i
A	0.23129		0.58663
B	0.11251	0.12132	0.51883
C	0.31877	0.03554	0.10031
D	0.27831	0.05926	0.17556
E	0.12070	0.11819	0.49475

Table 7
Calculation of the variation and index to the vector of the normalized reference ideal for the alternative *A*.

	C1	C2	C3	C4	C5	C6	
w_j	0.2262	0.2143	0.1786	0.1429	0.119	0.1190	$I_A^- = \sqrt{\sum_{j=1}^{n}(y'_{Aj})}^{2}$
y'_{Aj}	0.2262	0	0.1428	0.1429	0.119	0.0396	0.32823
${y'_{Aj}}^2$	0.0511	0	0.0204	0.0204	0.01416	0.0015	
$y'_{Aj} - w_j$	0	0.2143	0.0357	0	0	-0.0793	$I_A^+ = \sqrt{\sum_{j=1}^{n}(y'_{Aj} - w_j)^2}$
$(y'_{Aj} - w_j)^2$	0	0.0459	0.0012	0	0	0.0062	0.23129

Steps 4, 5, and 6 of the algorithm can be followed in Tables 5 and 6.
To show the calculation of the results obtained in Table 6, we proceed to make the calculation for the alternative *A*.

$$R_A = \frac{I_A^-}{I_A^+ + I_A^-} = \frac{0.23129}{0.32823+0.23129} = \frac{0.23129}{0.55952} = 0.586631$$

Step 7. Ranking of the alternatives.
Finally, we can say that the order of the candidates is given by $A > B > E > D > C$ (Table 7).

4.1. Sensitivity

We see that the difference between alternatives A and B, which appear at the top, is large enough so that a variation in the weights is unlikely to modify the results.

It can also be observed that the main difference between alternatives *B* and *E* lies in criteria C1, C2 and C5, hence we are going to study how a variation in the assessments of these criteria affects the results.

(a) Assume that the weight of C1 is diminished by 10%.
(b) Assume that the weight of C1 is diminished by 10% and the weight of C2 is raised by 10%.
(c) Assume that the weights of C1 and C5 are diminished by 10% and C2 is raised by 10%
(d) Assume that C1 is diminished by 10% and C2 is raised by 20%.
(e) Assume that C1 is diminished by 15% and C2 is raised by 15%.
(f) Assume that C1 and C2 are both diminished by 10% and C5 is raised by 20%.

Table 8
Results of R index in the sensitivity analysis for options a, b, c, d, e and f.

Alternatives	a	b	c	d	e	F
A	**0,57512981**	**0,56009002**	**0,55160178**	**0,53921675**	**0,53921675**	0,53060994
B	0,50958171	0,52084277	0,53016359	0,52894522	0,52894522	0,54738035
C	0,1724663	0,16637802	0,15827283	0,15369475	0,15369475	0,15007569
D	0,22174378	0,21350689	0,20502465	0,19774387	0,19774387	0,19396732
E	0,50133903	0,51609423	0,51641729	0,53559592	0,53559592	**0,53913841**

Table 9
Normalized original matrix and new alternative H.

Alternative	C1	C2	C3	C4	C5	C6
A	1	0	0.8	1	1	0.3333
B	0.8	0.9	0.9	1	0.5	0.3333
C	0.2857	0	0.7	0	1	0.3333
D	0.5714	0	0.5	1	1	0
E	0.6	1	0.8	0.5	1	1
H	**1**	**0.8**	**0.9**	**1**	**1**	**0.6667**

Table 10
Weighted normalized original matrix and new alternative H.

Alternative	C1	C2	C3	C4	C5	C6
A	0.22620	0	0.14288	0.14290	0.11900	0.03967
B	0.18096	0.19287	0.16074	0.14290	0.05950	0.03967
C	0.06463	0	0.12502	0	0.11900	0.03967
D	0.12926	0	0.08930	0.14290	0.11900	0
E	0.13572	0.21430	0.14288	0.07145	0.11900	0.11900
H	**0.2262**	**0.1714**	**0.1607**	**0.1429**	**0.1190**	**0.0793**

According to Table 8, we can see that as the weights of C1 and C5 diminish and C2 increases, the R value of *A* diminishes while *B* and *E* increases. However, no change in the alternatives ordering takes place until case (f), a situation which seems maybe too far from reality. Therefore, A is the best alternative.

5. RIM versus TOPSIS–VIKOR methods

5.1. Rank Reversal

When one or more alternatives are added or removed from a decision problem, the well-known Rank Reversal problem appears, resulting in an alteration of the ordering of some of the remaining alternatives.

According to [18] TOPSIS presents this problem. It can be solved by changing the norm $r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{j=1}^{m} x_{ij}^2}}$ for the norm $r_{ij} = \frac{x_{ij}}{Max_j\, x_{ij}}$. In addition to this, the introduction of two fictitious alternatives is required. The first alternative would correspond to the vector whose components represent the maximum of the domain for each criterion, while the other one corresponds to the minimum. In the case of the RIM method this is not necessary because the reference is implicit in definitions (1) and (2) by the range. Thus, we can affirm that the RIM method does not present Rank Reversal, considering that definition (2) works in relation to the range that is defined in advance and does not change if an alternative is added or eliminated. This means that if we modify the data of an alternative and/or we include or delete alternatives, then the rest of the alternatives do not modify their corresponding relative index.

Continuing with the previous example, let us suppose that we consider the inclusion of a new alternative, given by the vector H = [32, 8, 1, Good, Does not have, Normal]. Then, the normalized vector (Table 9) is independent of other alternatives and its corresponding values are [1, 0.8, 0.9, 1, 1, 0.6667]. The weighted normalized vector is [0.2262, 0.1714, 0.1607, 0.1429, 0.1190, 0.0793] as can be seen in Table 10 and finally, $I^+ = 0.06107$, $I^- = 0.14727$ and $R = 0.70687$ (Table 11).

The values of I^+, I^- and R of the rest of alternatives are not modified, i.e. they keep the same values as in Table 2 because the application of (2) is done for each alternative and criteria independently of the rest. The candidates are ordered in the following way, $H > A > B > E > D > C$. Likewise, if we eliminate an alternative the rest of the alternatives maintain their values for R and so the rank. Therefore, this method does not present Rank Reversal.

The main advantage of RIM versus TOPSIS and VIKOR is that the latter two methods are only applicable when the PIS and NIS coincide with the extremes of the range in which each one of the criteria is defined. Therefore, the above example cannot be solved by either TOPSIS or VIKOR.

Table 11
Indeces of original matrix and new alternative H.

Alternative	I_i^+	I_i^-	R_i
A	0.23129	0.32823	0.58663
B	0.11251	0.12132	0.51883
C	0.31877	0.03554	0.10031
D	0.27831	0.05926	0.17556
E	0.12070	0.11819	0.49475
H	**0.06107**	**0.14727**	**0.70687**

5.2. Relation between RIM, TOPSIS and VIKOR

When several decision making methods are applied to the same problem, it is well known that the results do not necessarily match. Generally this occurs when there are alternatives which are very close. On the contrary, if one alternative stands out, then one would expect that it is on the top of the final ranking.

Let us demonstrate how RIM is related to TOPSIS and VIKOR when the ideal solution coincides with the maximum or the minimum.

Theorem 1. Assume that M=[x_{ij}] is the matrix of data, $[A_j, B_j]$ the range for each criterion j, where $A_j < B_j$ and such that $A_j < x_{ij} < B_j$, $A_j = \min x_{ij\,j}$, $B_j = \max x_{ij\,j}$. If the ideal reference is $R_j = \max x_{ij\,j}$ then TOPSIS normalization is very close to the RIM method.

Proof:

Taking into account (2), it is possible to see that

$$f\left(x_{ij}, \max_j x_{ij}, \left[A_j, B_j\right]\right) = f\left(x_{ij}, B_j, \left[A_j, B_j\right]\right) = 1 - \frac{d_{\min}\left(x_{ij}, B_j\right)}{B_j - A_j} = 1 - \frac{B_j - x_{ij}}{B_j - A_j}$$

Furthermore, following the normalization given in [18] we obtain:

$$r_{ij} = \frac{x_{ij}}{\max x_{ij\,j}} = \frac{x_{ij}}{B_j}$$

Considering that, $\frac{x_{ij}}{B_j} + \varepsilon = 1 - \frac{B_j - x_{ij}}{B_j - A_j}$, then $\varepsilon = 1 - \frac{x_{ij}}{B_j} - \frac{B_j - x_{ij}}{B_j - A_j} = 1 - (\frac{x_{ij}}{B_j} + \frac{B_j - x_{ij}}{B_j - A_j})$ and $\lim_{x_{ij} \to B_j} \varepsilon = \lim_{x_{ij} \to B_j} (1 - (\frac{x_{ij}}{B_j} + \frac{B_j - x_{ij}}{B_j - A_j})) = 0$

Thus, it is possible to affirm that TOPSIS and RIM methods are very close when the value $x_{ij} \to \max x_{ij\,j}$.

Also, we see that $1 - \frac{B_j - x_{ij}}{B_j - A_j} = \frac{x_{ij} - \min x_{ij}}{\max x_{ij} - \min x_{ij}}$. This is the norm used in VIKOR.

The demonstration for the case in which the ideal corresponds to the minimum is similar and it is omitted. In this case we obtain that $1 - \frac{x_{ij} - A_j}{B_j - A_j} = \frac{Max\, x_{ij} - x_{ij}}{Max\, x_{ij} - Min\, x_{ij}}$ which coincides with the complementary norm of VIKOR.

The time complexity of these three methods is the same. Running time depends on the loops of steps 3, 4 and 5 of the proposed algorithms. In all cases, the order is quadratic, O(n∗m), and corresponds with the loops (for i=1,2,…m and j=1,2,…,n) aimed at exploring the decision matrix in all the cases. We understand that the only difference between them is the normalization function.

6. Conclusions

The MCDA constitutes a very useful tool of great utility to support the decision-making process. These methods have the purpose of facilitating the selection of the best alternatives through the evaluation of each alternative for a criteria set. Starting from the conception used by the TOPSIS and VIKOR methods regarding the identification of the ideal solution, we have obtained the RIM method that increases the possibilities of obtaining the best alternatives taking into account the reference ideal defined by the users. The RIM method is characterized by:

- The same conception of the TOPSIS and VIKOR methods with respect to the comparison of each alternative with the ideal solution. In our case, we define the reference ideal, and we introduce the variant where the reference ideal can be a set of values or a simple value that is between the maximum value and the minimum value, without having to be strictly the extreme values of each column (criterion) of the data.
- When the ideal values are at the extremes of the criteria domains, the RIM method has behavior similar to TOPSIS, further away from VIKOR.
- A better approach to changes in the data. Taking into account the expression (2), when we modify, include, or delete the information of an alternative it is only necessary to make the aggregation of the information for the new alternative.

The final assessment of the rest of the alternatives remains unchanged, differently from other methods in which it is necessary to process the information of all the alternatives again.

- Avoidance of Rank Reversal since the range does not vary and expression (2) measures the distance respect to it.

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